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RESEARCH PAPER

Evaluation of isotope-dilution mass spectrometry, internal standard, and external standard methods for PFAS quantification in tap water and soil

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Abstract
Accurate quantification of per- and polyfluoroalkyl substances (PFAS) in environmental matrices is crucial for reliable exposure assessment but remains challenging owing to low concentrations and matrix-induced variability. In this study, three calibration approaches, external standard (ES), internal standard (IS), and isotope dilution mass spectrometry (IDMS), were systematically evaluated for the determination of perfluorohexanesulfonic acid (PFHxS), perfluorooctanesulfonic acid (PFOS), perfluorooctanoic acid (PFOA), and perfluorononanoic acid (PFNA) in fortified tap water and fortified soil. Linearity of calibration curves was assessed across five concentration levels for each analyte. IDMS consistently demonstrated the highest linearity (coefficient of determination R^2 of 0.999) for all target analytes, followed by IS. ES displayed acceptable linearity, although the performance was slightly lower in soil matrices, with R^2 ranging from 0.992 to 0.996. Accuracy was evaluated using measured-to-assigned concentration ratios in spiked tap water. IDMS yielded ratios of 0.96–1.07 with small standard deviations, demonstrating high trueness. When IDMS was used as the reference, IS and ES showed matrix- and analyte-dependent biases. In tap water, IS improved agreement with IDMS relative to ES but still overestimated PFOS, PFOA, and PFNA, whereas ES showed the largest positive bias. In soil, IS produced values close to those of IDMS for all analytes, while ES exhibited greater variability and occasional underestimation. Overall, IDMS enabled the most consistent and reliable quantification across matrices. These results highlight the importance of isotope dilution calibration using structurally matched mass-labeled standards and indicate that IS and ES methods require careful consideration of matrix effects and surrogate selection.

Keywords Per- and polyfluoroalkyl substances (PFAS) · Isotope dilution mass spectrometry (IDMS) · Soil · Tap water · Internal standard method · External standard method

Introduction

For over half a century, per- and polyfluoroalkyl substances (PFAS) have been incorporated into a wide range of industrial and household products, including firefighting foams, antifouling textiles, non-stick cookware, and food-contact materials [1, 2]. The PFAS family includes thousands of individual compounds. Among these, perfluorohexanesulfonic acid (PFHxS), perfluorooctanesulfonic acid (PFOS), perfluorooctanoic acid (PFOA), and perfluorononanoic acid (PFNA) are particularly well-studied as they have been frequently detected in environmental matrices (e.g., water and soil) and prioritized in European food-related exposure and risk assessments. Notably, even though these compounds are typically detected at extremely low concentrations, often in the picogram-per-gram range or lower, their presence raises severe environmental and health concerns. This highlights the need for reliable and sensitive analytical approaches [3–5].

PFAS quantification is commonly performed using calibration curve-based approaches, often incorporating internal standard (IS) correction. In the IS method, structurally similar isotopically labeled analogues of the target analyte are typically introduced before sample extraction, enabling partial correction of recovery losses and matrix-induced

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suppression [6]. In contrast, external standard (ES) calibration relies on clean solvent-based calibration curves and does not account for variability introduced during extraction or ionization [7]. As the two approaches differ in concept and performance, the choice of calibration approach can considerably influence analytical outcomes. Isotope dilution mass spectrometry (IDMS), recognized as a primary method of measurement, is widely used for the characterization of certified reference materials owing to its traceability to the International System of Units and high metrological accuracy [8]. Although IDMS employs isotopically labeled compounds in a manner similar to the IS approach, it differs fundamentally in its calibration concept. In IDMS, structurally identical isotopically labeled analogues are added prior to extraction, and quantification is based directly on the ratio between native and labeled species. This ratio-based principle enables direct compensation of systematic bias and supports metrological traceability when gravimetric preparation is applied. In the present study, quantification based on isotope dilution was performed using ratio-based calculation according to the IDMS equation. Calibration curves were used solely to verify linear instrument response and not for mass fraction calculation. However, its application in routine PFAS monitoring studies is less common than calibration approaches based on external curves with internal standardization [9, 10]. In the present study, isotope dilution was implemented within a calibration-based liquid chromatography/tandem mass spectrometry (LC–MS/MS) framework rather than as a full primary measurement procedure, consistent with common targeted PFAS workflows. Several studies have demonstrated that IDMS can improve reproducibility and quantitative reliability, especially for complex matrices [11]. However, its widespread adoption is hindered by its time- and cost-intensive nature and the limited availability of suitable standards [12]. To the best of our knowledge, no study has directly compared IDMS, IS, and ES methods for PFAS quantification under uniform experimental conditions using both aqueous and solid environmental matrices. Although IDMS has been shown to improve accuracy in certain scenarios, its effectiveness may depend on factors such as sample matrix, analyte concentration, or extraction procedure, and therefore may not be superior across all analytical situations [11, 13, 14].

To address this knowledge gap, this study was aimed at systematically comparing three quantification approaches, i.e., IDMS, IS, and ES methods, for PFHxS, PFOS, PFOA, and PFNA. The selected analytes represent priority legacy PFAS that are widely monitored, frequently detected in environmental and food-related studies, and analytically well characterized. Their inclusion provides a robust basis for evaluating the performance of calibration strategies across sulfonic acid and carboxylic acid classes and varying chain lengths. The evaluation was performed using fortified tap water and fortified soil samples, selected to represent typical environmental matrices targeted in PFAS monitoring programs worldwide. The aim of this study was not to reproduce specific routine LC–MS/MS quantification workflows commonly applied in practice, but to compare calibration concepts under controlled experimental conditions to evaluate method-dependent bias and analytical robustness.

Materials and methods

Chemicals and reagents

Individual native PFAS standards and mass-labeled PFAS standards were purchased from Wellington Laboratories Inc. (Guelph, ON, Canada). These standards included PFHxS, PFOS, PFOA, PFNA, sodium perfluoro-1-[1,2,3-$^{13}C_3$] hexane sulfonic acid ($^{13}C_3$-PFHxS), sodium perfluoro-1-hexane [$^{18}O_2$] sulfonic acid ($^{18}O_2$-PFHxS), sodium perfluoro-1-[1,2,3,4-$^{13}C_4$]-octane sulfonic acid ($^{13}C_4$-PFOS), perfluoro-*n*-[1,2,3,4-$^{13}C_4$] octanoic acid ($^{13}C_4$-PFOA), perfluoro-*n*-[1,2,3,4,5-$^{13}C_5$] nonanoic acid ($^{13}C_5$-PFNA), and *N*-methyl-d_3-perfluoro-1-octanesulfonamide (d_3-*N*-MeFOSA). All standards had a concentration of 50 µg/mL and a purity > 98%. Ultrapure water (PFOS, PFOA analysis grade), methanol (> 99.8%; pesticide residue-PCB analytical grade), acetic acid (> 99.7%; guaranteed reagent), ammonia solution (25%; Wako 1 st grade), and ammonium acetate (> 97%; guaranteed reagent) were obtained from FUJIFILM Wako Pure Chemical Corporation (Osaka, Japan). Weak anion-exchange (WAX) solid-phase extraction (SPE) cartridges (Oasis® WAX for PFAS analysis; 6 mL, 150 mg, 30 µm particle size) were supplied by Waters Co. (Milford, MA, USA).

Preparation of standard solutions

All native and mass-labeled standard solutions were prepared gravimetrically. Stock solutions of PFHxS (1.0 µg/g), PFOS (10 µg/g), PFOA (0.10 µg/g), and PFNA (0.10 µg/g), as well as their corresponding mass-labeled internal standards for IDMS, $^{13}C_3$-PFHxS, $^{13}C_4$-PFOS, $^{13}C_4$-PFOA, and $^{13}C_5$-PFNA were prepared by diluting each original solution with methanol. Stock solutions of $^{18}O_2$-PFHxS (10 µg/g), used as a common internal standard for IS-based quantification, and d_3-*N*-MeFOSA (10 µg/g), used as a syringe spike, were prepared using the same protocol described above. The mass-labeled internal standards used for the IDMS and IS methods are summarized in Table S1 of the Supplementary Information (SI). Working mixture standards were prepared by further diluting the stock solutions. The native mixture solution contained 34 ng/g PFHxS, 448 ng/g PFOS, 7.7 ng/g PFOA, and 3.7 ng/g PFNA. The mass-labeled mixture solution contained 34 ng/g $^{13}C_3$-PFHxS, 448 ng/g $^{13}C_4$-PFOS, 7.7 ng/g $^{13}C_4$-PFOA, 3.7 ng/g

$^{13}C_5$-PFNA, and 15 ng/g $^{18}O_2$-PFHxS. Additionally, a syringe spike solution containing 15 ng/g of d_3-*N*-MeFOSA was prepared. The syringe spike was used solely to monitor injection stability and short-term instrumental variability across replicate injections. For IDMS, a calibration mixture was prepared such that the ratio of native PFAS to mass-labeled PFAS corresponded to that of the sample extract. The calibration curve standards were gravimetrically prepared to achieve peak area ratios of 0, 0.5, 1, 1.5, and 2 between the target analytes and their corresponding mass-labeled PFAS; the objective was to verify the linear relationship between peak area ratios and mass ratios. Unweighted linear regression was applied uniformly to all calibration approaches to enable direct comparison under identical regression conditions.

Sample collection and preparation

Tap water samples were collected directly from a laboratory faucet in Tsukuba, Ibaraki, Japan, in March 2024. Samples were collected in 500 mL polypropylene (PP) bottles after allowing the tap to run for 1 min. The working mixture standard containing PFHxS, PFOS, PFOA, and PFNA was spiked into the samples at amounts of 5 ng, 77 ng, 1.2 ng, and 0.6 ng, respectively, to achieve mass fractions equivalent to those in the tap water sample extracts. Immediately after spiking with the native standards, the samples were subjected to extraction.

The soil sample consisted of 70% commercially available potting soil, 20% dried andosol, and 10% dried lowland soil. The potting soil was intentionally exposed to PFAS-contaminated water and subsequently dried. The soil was homogenized using a soil mixing machine for 6 h, after which the particle size was regulated using a 2 mm sieve. The sample was stored in a light-tight plastic bottle and kept in a freezer at −25 °C until analysis.

Sample pretreatment and instrumental analysis

Five hundred milliliters of tap water spiked with 0.6–77 ng of mass-labeled standards were subjected to SPE using Oasis® WAX cartridges. The extraction procedure followed a previously published method [15]. Briefly, the cartridges were preconditioned sequentially with 4 mL of 0.1% (*v*/*v*) ammonia in methanol, 4 mL of methanol, and 4 mL of ultrapure water, followed by loading of 500 mL of the tap water sample. After washing with 5 mL of ultrapure water and 4 mL of acetate buffer solution, the cartridges were centrifuged at 3000 rpm for 2 min to ensure drying. The target analytes were subsequently eluted using methanol and 0.1% (*v*/*v*) ammonia in methanol. To minimize potential adsorption of PFAS onto sample container walls, the container was rinsed with methanol during the elution step, and the rinse was included in the elution, following the procedure described in ISO 21675 [15]. Both eluates were concentrated under a nitrogen stream at 40 °C to a final volume of 0.5 mL. Thereafter, 2.4 ng of d_3-*N*-MeFOSA was added as a syringe spike to the concentrate, and the volume was adjusted to 0.5 mL under the same conditions.

Approximately 5 g of the air-dried soil sample was weighed into a 50 mL PP centrifuge tube and spiked with 0.6–77 ng of mass-labeled standards. The extraction and purification procedure were performed according to a previously reported method [16]. Briefly, solid–liquid extraction was performed under horizontal shaking with 10 mL of 0.5% (*v*/*v*) ammonia in methanol solution at 300 rpm for 30 min. The sample was then placed in an ultrasonic bath at room temperature for 10 min and centrifuged at 3000 rpm for 15 min. The supernatant was transferred to a new 50 mL PP centrifuge tube. This extraction process was repeated twice, after which the combined supernatant was centrifuged again and transferred to a new centrifuge tube to completely separate the soil residue from the extract. The extract was concentrated under a gentle stream of nitrogen gas at 40 °C to approximately 5 mL, diluted with 20 mL of ultrapure water, and adjusted to pH 3 using acetic acid. Clean up was then performed via SPE using Oasis® WAX cartridges, following the same procedure as that for tap water samples. The tap water and soil samples were analyzed in triplicate.

Aliquots of 10 µL were injected into a 20 A high-performance liquid chromatography (HPLC) system (Shimadzu, Kyoto, Japan) coupled with an 8060NX triple quadrupole tandem mass spectrometer (Shimadzu, Kyoto, Japan) operating in the negative electrospray ionization mode with multiple reaction monitoring. Chromatographic separation of analytes was achieved using a ZORBAX Eclipse XDB-C8 guard column (2.1 mm i.d. × 12.5 mm length, 5 µm particle size; Agilent Technologies, Foster City, CA), and BetaSil® C18 analytical column (2.1 mm i.d. × 50 mm length, 5 µm particle size; Thermo Fisher Scientific, Waltham, MA, USA). To minimize background contamination originating from the HPLC system, an InfinityLab PFC Delay Column (4.6 mm i.d. × 30 mm length; Agilent Technologies, Santa Clara, CA, USA) was installed. Details of chromatographic conditions and MS/MS parameters are presented in Tables S2 and S3.

Calculation of mass fractions using IDMS

The mass fractions of PFHxS, PFOS, PFOA, and PFNA were quantified using IDMS via one-point calibration with isotope ratio matching, according to the following expression:

$$w_x = \frac{M_{cal.} \times w_{cal.} \times M_{surr.} \times R_{sample}}{M_{surr.cal.} \times M_{sample} \times R_{cal.}}$$

where w_x is the mass fraction of the target analyte in the sample; $w_{cal.}$ is the mass fraction of the target analyte in the native mixture solution for IDMS; $M_{cal.}$ is the mass of the native mixture solution in the calibration mixture solution for IDMS; $M_{surr.}$ and $M_{surr.cal.}$ are the masses of the mass-labeled mixture solution added to the samples and calibration mixture solution for IDMS, respectively; M_{sample} is the sample mass; R_{sample} is the peak area ratio of the target analyte to its corresponding mass-labeled compound in the sample extract; and $R_{cal.}$ is the corresponding ratio in the calibration mixture solution.

Quality assurance and quality control

Ultrapure water and a mixture of the soil extraction solvent and ultrapure water as procedural blanks were extracted in the same manner as the samples, with all analytes below the method quantification limits (MQLs). The MQLs, determined at a signal-to-noise ratio of 10, ranged from 22 pg/g to 219 pg/g. The recoveries of $^{13}C_3$-PFHxS, $^{13}C_4$-PFOS, $^{13}C_4$-PFOA, $^{13}C_5$-PFNA, and $^{18}O_2$-PFHxS in tap water were 82.5% ± 8.9%, 94.3% ± 13.1%, 85.8% ± 10.6%, 89.6% ± 13.4%, and 88.4% ± 15.0%, respectively. For soil samples, the recoveries of $^{13}C_3$-PFHxS, $^{13}C_4$-PFOS, $^{13}C_4$-PFOA, $^{13}C_5$-PFNA, and $^{18}O_2$-PFHxS were 101.6% ± 0.8%, 104.6% ± 1.7%, 72.9% ± 3.5%, 79.7% ± 5.7%, and 86.3% ± 7.4%, respectively.

Statistical analysis

Linear regression analysis was performed using Microsoft Excel (Microsoft Corporation, Redmond, WA, USA) to generate calibration curves. The linearity of the calibration curves was assessed using lack-of-fit (LOF) tests to identify significant deviations from linearity. The LOF tests were performed in R software (version 3.6.1; R Foundation for Statistical Computing, Vienna, Austria). The normality of residuals was evaluated using the Shapiro–Wilk test, conducted in GraphPad Prism 10 (version 10.4.2; GraphPad Software, San Diego, CA, USA). Homoscedasticity was assessed using either the Breusch–Pagan test or White's test implemented in Python (version 3.11.8; Python Software Foundation). The test was selected based on the result of the normality test. *p*-values less than 0.05 were considered statistically significant.

Results and discussion

Linearity assessment of calibration curves

The linearity of calibration curves was evaluated for the IDMS, IS, and ES methods. Regression parameters, including the coefficient of determination (R^2), slope, intercept, standard errors (SEs), *p*-values, and *t*-values, were analyzed, in combination with residual analysis and homoscedasticity assessments via the Breusch–Pagan, White, and LOF tests. As shown in Table 1, all target analytes quantified using IDMS exhibited consistently high linearity, with R^2 values of 0.999. The IS method also demonstrated high linearity (0.986–0.999), although slightly larger deviations were observed compared with IDMS, particularly for compounds such as PFOA and PFNA, for which the internal standard was not structurally well-matched (e.g., $^{18}O_2$-PFHxS used as a surrogate for both PFOA and PFNA). The ES method showed the lowest R^2 values, particularly for PFHxS (0.987) and PFOA (0.984). Although R^2 alone is insufficient to establish linearity [17], values above 0.98 are often considered indicative of acceptable linearity. Compared with IDMS, the IS method displayed greater variability in slopes and intercepts. *p*- and *t*-values were used to support the statistical validity of the regression slopes. For all methods, the regression slopes were statistically significant ($p < 0.001$), indicating a significant difference from zero [7]. *t*-values derived from the regression coefficients provided insights into the reliability of the slopes and intercepts [18]. IDMS resulted in the highest *t*-values (47.6–53.0), followed by IS (14.3–50.0) and ES (13.5–24.7), reflecting stronger statistical certainty in the regression coefficients obtained by IDMS [19]. Moreover, IDMS consistently displayed lower SE values for the slopes compared with the IS method (e.g., IDMS: 0.049 vs. IS: 0.084 for PFOS; and IDMS: 0.038 vs. IS: 0.45 for PFNA), although it exhibited a slightly higher SE for PFHxS. Overall, these results indicated that IDMS-based quantification results in superior precision [20]. The ES method displayed the lowest *t*-values, raising concerns regarding the precision and statistical validity of slopes. These outcomes imply the weakest confidence in ES-based calibration, particularly at lower concentrations.

In addition to the basic regression metrics, residual plots were generated (Fig. S1) to verify the suitability of the linear regression models. Across all quantification methods, the residuals were randomly scattered, and no specific trend was observed. However, for the ES method, PFOS and PFNA showed greater dispersion of residuals at lower concentration levels. Heteroscedasticity was evaluated using either the Breusch–Pagan or White test to assess the assumption of constant variance of the residuals, an important requirement for the validity of linear regression [21]. The resulting *p*-values are summarized in Table S4. IDMS showed high *p*-values (e.g., 0.896 for PFNA and 0.569 for PFOS), indicating no significant heteroscedasticity. The IS method also demonstrated high *p*-values across most analytes (e.g., 0.436 for PFOA and 0.394 for PFOS), supporting the assumption of homoscedasticity. These results implied that both IDMS and IS produce calibration

Table 1 Linear regression results for calibration curves obtained using IDMS, IS, and ES methods

		PFHxS	PFOS	PFOA	PFNA
IDMS	Slope ± SE	4.15 ± 0.087	2.90 ± 0.049	13.4 ± 0.29	2.00 ± 0.038
	Intercept ± SE	0.0734 ± 0.10	0.0219 ± 0.059	0.199 ± 0.24	1.90×10^{-4} ± 0.046
	R^2	0.999	0.999	0.999	0.999
	p-value	2.04×10^{-5}	1.09×10^{-5}	2.15×10^{-5}	1.48×10^{-5}
	t-value	47.6	58.7	46.8	53.0
IS	Slope ± SE	2.59 ± 0.052	1.20 ± 0.084	14.1 ± 0.36	7.09 ± 0.45
	Intercept ± SE	0.0971 ± 0.15	4.62 ± 3.5	0.00532 ± 0.20	0.168 ± 0.13
	R^2	0.999	0.986	0.998	0.988
	p-value	1.76×10^{-5}	7.35×10^{-5}	3.53×10^{-5}	5.54×10^{-4}
	t-value	50.0	14.3	39.7	15.8
ES	Slope ± SE	3.79×10^{8} ± 2.5×10^{7}	1.75×10^{8} ± 2.5×10^{6}	2.06×10^{9} ± 1.5×10^{8}	1.03×10^{9} ± 4.2×10^{7}
	Intercept ± SE	-1.10×10^{5} ± 3.2×10^{5}	1.68×10^{6} ± 1.4×10^{6}	-1.84×10^{5} ± 4.0×10^{5}	5.69×10^{4} ± 5.3×10^{4}
	R^2	0.987	0.994	0.984	0.995
	p-value	6.05×10^{-4}	4.73×10^{-4}	8.75×10^{-4}	1.46×10^{-4}
	t-value	15.3	23.3	13.5	24.7

SE, standard error

curves in which the variance of the residuals remains stable over the concentration range, supporting the validity of linear regression. In contrast, the ES method exhibited the lowest *p*-values, particularly for PFHxS (0.052) and PFOS (0.096), suggesting potential heteroscedasticity. This variability in residual variance may indicate a deviation from linearity over the calibration range, especially at extremely low or high concentrations, which is critical for trace-level quantification. Therefore, although all three methods produced high R^2 values (above 0.98), the heteroscedasticity results highlighted that the linear models derived from IDMS and IS are more statistically robust. The heteroscedasticity observed in the ES model may reflect matrix-induced ionization variability or concentration-dependent extraction efficiency.

To further evaluate the sufficiency of the linear models, LOF tests were performed [22], and the corresponding *p*-values are listed in Table S5. Among the three quantification methods, only IDMS yielded non-significant LOF *p*-values across all target analytes, indicating no statistically significant deviation from linearity [23]. In contrast, the IS and ES methods exhibited significant LOF in several cases ($p < 0.05$), raising concerns regarding the linearity of these models under the considered experimental conditions and highlighting the need for alternate regression models [24, 25]. This LOF may result from nonlinear behavior in the response at extremely low or high concentrations or uncorrected matrix effects in ES and IS calibrations. Although both IS and ES methods may still be acceptable for practical applications, the results of this work clearly demonstrate that IDMS provides more consistent and statistically robust quantification. This emphasizes the importance of using mass-labeled compounds of target analytes as internal standard for reliable measurements.

Evaluation of matrix effects using matrix-matched calibration

To evaluate the matrix effects associated with each quantification method, matrix-matched calibration solutions were prepared using matrix extracts processed in the same manner as the quantification samples. The matrix solution was generated using the residues of extracted tap water and soil samples, as PFAS-free tap water and soil are challenging to obtain. The absence of target analytes was confirmed before use. Matrix-matched calibration solutions were prepared using extracts obtained from the same matrices after undergoing the sample preparation procedure. This approach provides an estimate of residual ionization effects in the final extracts used for quantification. However, because some co-extracted components may already have been removed during the initial extraction and cleanup, the magnitude of matrix effects may be lower than that observed in untreated matrices. Calibration standards were spiked into these matrix solutions to construct matrix-matched calibration curves. Matrix effects were estimated by comparing the slopes of

matrix-matched and solvent-based calibration curves, as described elsewhere [26]. The matrix effect was expressed as a percentage slope ratio, as summarized in Table 2.

In tap water, all quantification methods exhibited slope ratios within the range of 4.71%–7.95%, indicating mild ion enhancement. The ES method showed slightly higher ratios than those for the IS and IDMS methods, reflecting its greater sensitivity to matrix-induced signal enhancement. In contrast, matrix effects in soil extracts were consistently below 1% for all compounds and methods, with values ranging from 0.73% to 0.92%. These results suggest minimal ion enhancement, likely owing to the effective removal of interferences during sample purification. According to previously reported classification criteria [27, 28], all values obtained in this study fall within the category of "soft" matrix effects. The matrix effects in soil were particularly low, indicating negligible interference. Notably, matrix effects reported for PFAS in existing studies vary widely depending on the sample type and analytical conditions. For example, Keršňáková et al. reported slope-ratio matrix factors ranging from approximately −12.3% to 16.6% for PFAS in groundwater, interpreted as weak matrix effects [29]. In contrast, Tang et al. reported considerably larger effects (56.8%–120.0%) for PFAS in industrial wastewater, highlighting that matrix complexity can strongly affect analytical accuracy [30]. In the present study, the matrix effects were 4.71%–7.95% in tap water and < 1% in soil for all compounds and quantification methods. These values reflect comparatively lower matrix effects compared with those reported previously, indicating that the applied sample purification process successfully minimized matrix-related interferences.

The present matrices represent relatively controlled and purified conditions. In more matrix-affected samples, such as sewage sludge, untreated wastewater, or complex soil extracts, substantially larger and more variable matrix effects have been reported in PFAS analysis. For example, ion suppression or enhancement ranging from −55% to 42% has been observed in complex groundwater matrices for representative PFAS [29]. Previous studies have also demonstrated that the magnitude of matrix effects strongly depends on matrix composition and extraction/cleanup method [31–33]. Under such conditions, calibration methods that rely on surrogate correction (IS) or external calibration (ES) may be more susceptible to analyte- and matrix-dependent bias, whereas compound-specific isotope dilution is expected to better compensate for ionization variability. Therefore, the differences observed in this study may represent conservative estimates of method-dependent bias under moderate matrix influence.

Table 2 Matrix effects (%) expressed as slope ratios for four PFAS in tap water and soil, derived using IDMS, IS, and ES methods

	PFHxS	PFOS	PFOA	PFNA
Tap water				
IDMS	4.76	4.71	4.74	4.78
IS	4.76	4.84	4.75	4.82
ES	5.45	5.53	7.95	5.28
Soil				
IDMS	0.80	0.80	0.76	0.73
IS	0.83	0.79	0.85	0.77
ES	0.89	0.80	0.92	0.80

Accuracy of IDMS for PFAS quantification in tap water

The accuracy of IDMS was evaluated by comparing the measured mass fractions of PFHxS, PFOS, PFOA, and PFNA in spiked tap water samples with their gravimetrically assigned values. Mass fractions of PFHxS, PFOS, PFOA, and PFNA obtained by IDMS were 8.13 pg/g ± 0.10 pg/g, 105.89 pg/g ± 1.86 pg/g, 2.29 pg/g ± 0.002 pg/g, and 0.89 pg/g ± 0.02 pg/g, respectively, corresponding to measured-to-assigned ratios of 1.07 ± 0.014, 0.96 ± 0.017, 0.98 ± 0.023, and 1.020 ± 0.030 (Fig. 1). The reported intervals represent the standard deviation of replicate measurements ($n = 3$). Notably, the ratios ranged from 0.96 to 1.07, with small standard deviations (≤ 0.03), indicating only minor deviations from the gravimetrically assigned values and no clear bias across all analytes. Together with the high linearity ($R^2 = 0.999$), small SEs, and absence of significant heteroscedasticity and LOF (Table 1), these results support the use of IDMS as a reference method for subsequent comparisons with the IS and ES methods.

Comparison of IS and ES methods with IDMS in tap water

Using IDMS measurements as the reference, Fig. 2 shows the relative mass fraction of PFHxS, PFOS, PFOA, and PFNA obtained by the IS and ES methods in tap water samples. The IS method yielded results close to those obtained using IDMS for PFHxS (0.93 ± 0.022), PFOA (1.07 ± 0.010), and PFNA (1.07 ± 0.012), although the mass fraction for PFOS (1.14 ± 0.032) was moderately overestimated. In contrast, the ES method consistently yielded higher values, especially for PFOS (1.16 ± 0.010) and PFNA (1.14 ± 0.026). These results indicate that the IS method provides better agreement with IDMS than the ES method. The consistent overestimation observed with the ES method reflects its limited ability to correct matrix effects and instrumental fluctuations [34, 35]. Because the magnitude of the observed overestimation exceeds the contribution expected from matrix effects alone, additional factors, such as surrogate mismatch, differences in recovery behavior during

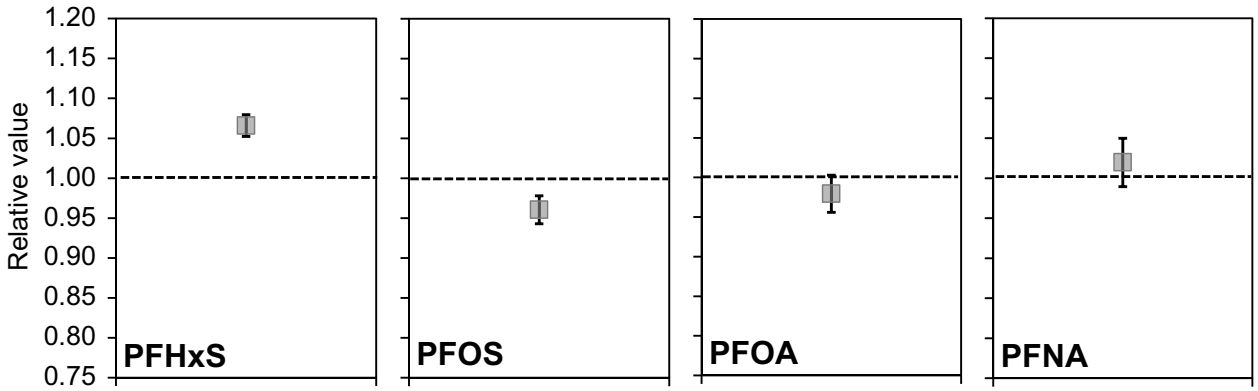


Fig. 1 Concentrations of PFHxS, PFOS, PFOA, and PFNA derived by IDMS relative to the spiked concentrations. Dashed lines indicate relative values of spiked concentrations (1.0), and error bars indicate standard deviation ($n = 3$)

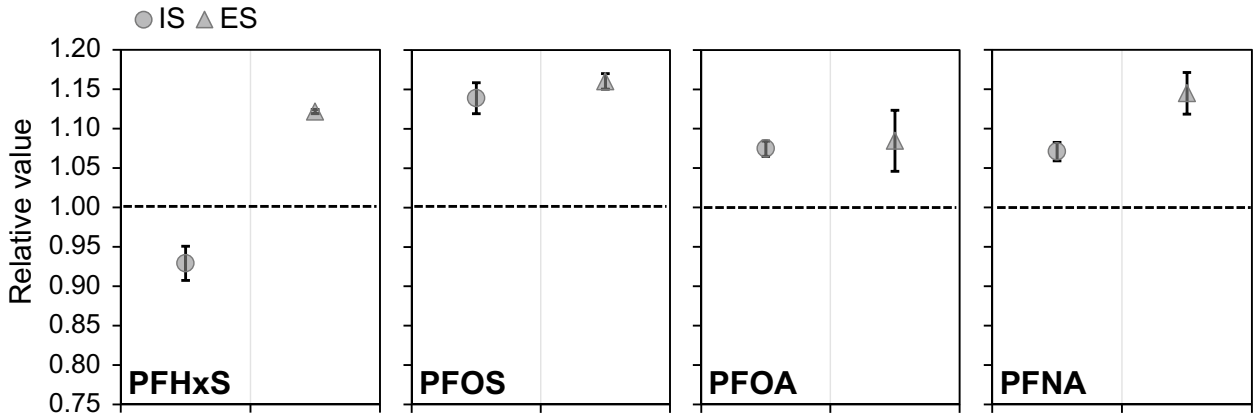


Fig. 2 Relative mass fractions of PFHxS, PFOS, PFOA, and PFNA in tap water determined by IS and ES methods, compared with those extracted using IDMS. Dashed lines indicate relative values of IDMS measurements (1.0), and error bars represent standard deviation ($n = 3$)

sample preparation, or concentration-related uncertainties, likely contributed to the observed bias. Overall, although the IS method did not fully correct for the bias, it demonstrated closer agreement with IDMS relative to the ES method. The remaining analyte-specific deviations in IS results, such as for PFOS and PFNA, may be attributable to structural dissimilarities between the target analyte and internal standard (e.g., PFOS against $^{18}O_2$-PFHxS), such as differences in carbon chain length and terminal functional groups [36]. It should be noted that the choice of labeled internal standards may influence quantitative agreement. Alternative isotopically labeled standards with closer structural similarity to the target analytes (e.g., fully ^{13}C-labeled analogues) could potentially reduce analyte-specific deviations. The present comparison was performed using consistent surrogate selection across methods to enable systematic evaluation rather than to optimize surrogate choice. Similar observations have been reported by Taniyasu et al. [37] in an interlaboratory trial, which revealed that the use of internal standards not isotopically identical to the target PFAS could result in either underestimation or overestimation of their concentrations. Taniyasu et al. [37] thus emphasized the importance of selecting structurally matched isotope-labeled surrogates to ensure accurate quantification.

In addition to surrogate-related biases, both the IS and ES methods displayed a tendency toward overestimation, particularly for PFOS and PFNA (IS) and PFHxS, PFOS, and PFNA (ES). The IS method yielded measured-to-assigned ratios of 1.09 ± 0.0010 for PFOS and 1.09 ± 0.024 for PFNA, whereas the ES method resulted in even greater overestimation: PFHxS: 1.20 ± 0.018, PFOS: 1.11 ± 0.029, and PFNA: 1.17 ± 0.039. These results indicate that methods lacking internal standard correction are more sensitive to matrix effects, even in simple matrices like tap water. After concentration of the tap water extract to 0.5 mL, fine white suspended particulates were visually observed in the final extract. Although their chemical identity was not characterized in this study, these particulates likely represent residual co-extracted matrix components that remain after sample preparation. Such co-extractives may influence

electrospray ionization by altering droplet surface activity or charge competition effects, potentially leading to ion suppression or enhancement. Because these components would be present in all extracts, including those analyzed by IDMS, the observed differences between calibration approaches are unlikely to arise solely from their presence. Instead, the ratio-based nature of IDMS may reduce the impact of common ionization changes, whereas ES and surrogate-based IS approaches may remain more sensitive to matrix-dependent variability.

Comparison of IS and ES methods with IDMS in soil

Figure 3 presents the relative values for soil, obtained using the IS and ES methods with IDMS as reference. In contrast to the tap water samples, the final concentration volume for soil extracts was 1.0 mL, and no visible suspended particulates were observed after evaporation. For soil samples, direct comparison with nominal spiked amounts was not applicable because the samples were prepared by exposure to PFAS-contaminated water. Based on IDMS measurements, the mass fractions in soil were 1.69 ng/g ± 0.09 ng/g for PFHxS, 14.20 ng/g ± 0.18 ng/g for PFOS, 0.40 ng/g ± 0.01 ng/g for PFOA, and 0.18 ng/g ± 0.005 ng/g for PFNA. These values were used as references for evaluating the IS and ES methods. The IS method yielded values close to 1.00 for all analytes: PFHxS (0.95 ± 0.032), PFOS (0.96 ± 0.041), PFOA (0.91 ± 0.039), and PFNA (0.99 ± 0.041). In contrast, the ES method showed greater variability and occasional underestimation (e.g., PFOS: 0.82 ± 0.051), likely reflecting the absence of internal standard correction and higher sensitivity of ES calibration to minor matrix-related and instrumental variations. Interestingly, the deviations of both IS and ES values from IDMS values were smaller in soil than in tap water. This may be attributable to the differences in final concentration volume and co-eluted matrix components.

Together, the findings from tap water and soil matrices highlight not only the differences in analytical performance but also practical considerations when selecting a quantification strategy. IDMS requires isotopically identical labeled standards for each analyte and careful control of isotope ratios, which may increase cost, preparation effort, and dependency on standard availability. In contrast, IS and ES approaches can be implemented with fewer labeled compounds and are therefore more accessible for routine laboratories. Nevertheless, the present results indicate that such simplified calibration strategies require careful surrogate selection and matrix-specific validation to avoid method-dependent bias.

Conclusion

Three quantification approaches, IDMS, IS, and ES, were evaluated for the determination of PFHxS, PFOS, PFOA, and PFNA in tap water and soil. IDMS provided calibration curves with high linearity, low SEs, and no significant heteroscedasticity or LOF, and yielded measured-to-assigned ratios close to the gravimetrically assigned values in spiked tap water, demonstrating high trueness. Matrix-matched calibration revealed mild ion enhancement in tap water and negligible matrix effects in soil; however, the IS and ES methods still exhibited analyte- and matrix-dependent biases when compared with IDMS. In tap water, the IS method improved agreement with IDMS compared with the ES method, although overestimation persisted for PFOS, PFOA, and PFNA. The ES method was noted to be the most sensitive to matrix effects and instrumental variability. In soil, the IS method provided values close to IDMS values, while ES still showed larger variability. Overall, IDMS delivered the most consistent and reliable quantification across matrices. Notably, optimal IDMS performance is achieved when the

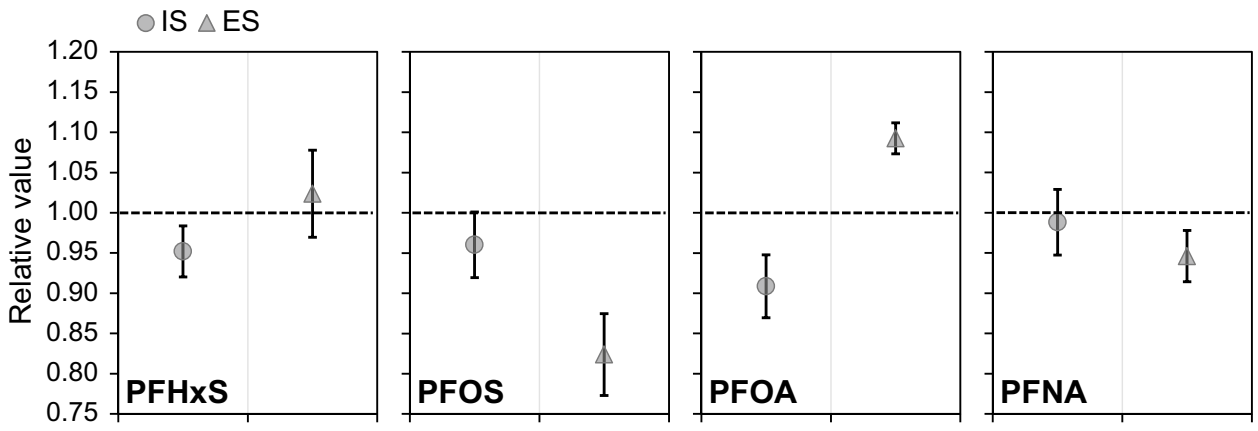


Fig. 3 Relative mass fractions of PFHxS, PFOS, PFOA, and PFNA in soil determined by IS and ES methods, compared with those extracted using IDMS. Dashed lines indicate relative values of IDMS measurements (1.0) and error bars represent standard deviation ($n = 3$)

concentration of the labeled spike is comparable to that of the native analyte, minimizing uncertainty in signal ratios. In this study, native-to-labeled ratios were controlled within a defined range (0–2), and calibration linearity was verified without significant LOF. Therefore, quantification was performed within the validated linear interval. In environmental samples with highly variable and unknown concentrations, appropriate adjustment of spike levels remains essential for ensuring IDMS performance. These results highlight that, while isotope dilution calibration with structurally matched mass-labeled standards enables highly robust quantification, residual matrix effects may persist. Consequently, careful surrogate selection and matrix-specific evaluation remain essential for the reliable interpretation of PFAS measurements when IS and ES approaches are applied.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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